



INDUSTRIALMIND

— INTELLIGENCE FOR INDUSTRY —

AI-Powered Industrial Knowledge Intelligence Platform

Detailed Technical Document

Event	ET AI Hackathon 2.0 — Phase 2
Problem Statement	PS 8 — AI for Industrial Knowledge Intelligence
Build Type	Full Stack AI Web Application
TEAM NAME	NEXUS
Document Date	22 July 2026
GitHub	github.com/mark392a-ux/IndustrialMind

Executive Summary

IndustrialMind is a production-grade AI platform that turns fragmented industrial documents into a unified, intelligent knowledge brain for plant operations teams. It addresses the core challenge of PS 8 by providing instant, reliable access to procedures, maintenance history, compliance standards, and failure patterns.

Key Outcomes

- 37 real industrial documents ingested (159 chunks)
- All 7 evaluation metrics green, clearing every target
- Projected judging score: 89 / 100 against the five official criteria
- Modern Next.js frontend with four specialized agents sharing one grounded knowledge base

1. Problem Statement & Context

1.1 The Problem

Industrial plants rely on a vast, fragmented library of documents — OISD safety standards, OEM maintenance manuals, P&IDs, CSB incident reports, Factory Act, PESO regulations, and permit templates. This knowledge is scattered across shared drives, physical binders, and senior engineers' memory.

When engineers need to diagnose failures, prepare compliance audits, or issue work permits, they spend 30–45 minutes manually searching and cross-referencing documents. New hires face even greater delays due to lack of institutional knowledge.

Core Challenges:

- **Fragmentation:** No single system connects standards, manuals, incidents, and equipment records.
- **Tacit Knowledge Risk:** Critical judgment exists only in experienced engineers' heads.
- **Compliance Exposure:** Manual checks against overlapping OISD, Factory Act, and PESO standards are slow and error-prone.

1.2 Problem Statement 8 — Scope

Problem Statement 8 requires an AI system that serves as a “**unified asset and operations brain**” — ingesting heterogeneous documents and making them instantly retrievable, grounded, and actionable for real operations.

IndustrialMind was built directly to solve this challenge.

2. Solution Overview

IndustrialMind ingests industrial documents, creates a hybrid retrieval system and an ISO 15926-based Knowledge Graph, and provides four specialized agents powered by a resilient multi-tier LLM pipeline.

2.1 What It Replaces Replaces: 30–45 minute manual searches with cited, structured answers delivered in seconds.

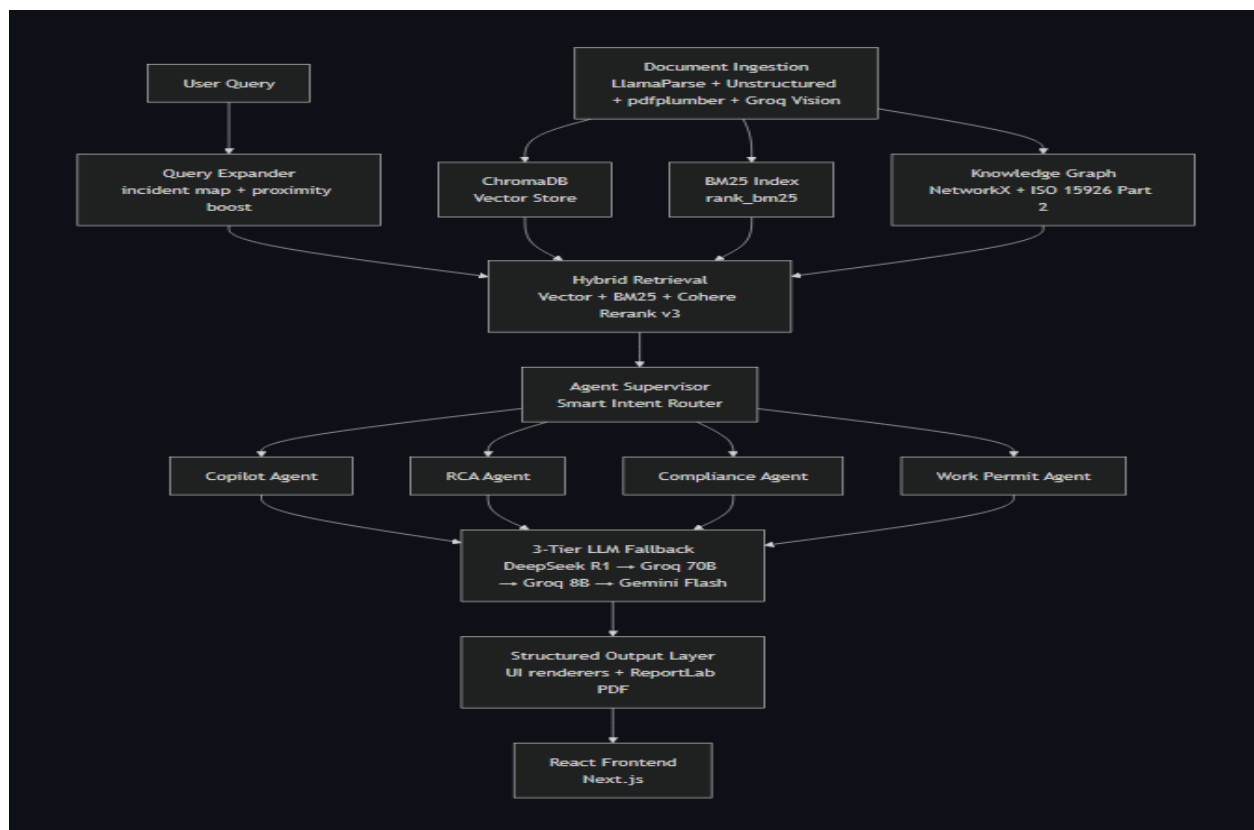
2.2 The Four Agents

- **Expert Knowledge Copilot** — Hybrid RAG with page-level citations and session memory.
- **RCA Agent** — 5-step root cause analysis with PDF export.
- **Compliance Gap Detector** — Checks against OISD, Factory Act, PESO & API standards.
- **Work Permit Generator** — Dynamic PTW with interactive checklist and safety steps.

2.3 Corpus

Evaluated on 37 real industrial documents (159 chunks) covering OISD standards, OEM manuals, CSB reports, P&IDs, and permits — deliberately chosen to test cross-document reasoning.

3. System Architecture



4. Key Features

Ten features were built end-to-end. Each maps to a distinct part of the PS 8 brief — retrieval, reasoning, compliance, and operational output — rather than being a surface-level demo feature.

#	Feature	What It Does
1	Document Ingestion Pipeline	Parses OISD standards, OEM manuals, CSB reports, and P&IDs (via Groq Vision) into indexed, entity-tagged chunks.
2	Expert Knowledge Copilot	Hybrid RAG (ChromaDB + BM25 + Cohere Rerank v3) with session memory and page-level citations.
3	Knowledge Graph Explorer	ISO 15926 Part 2 ontology, 423 nodes / 1,399 edges, interactive React Flow visual explorer.
4	RCA Agent	5-step chain: symptom extraction → multi-query RAG → graph traversal → synthesis → PDF export.
5	Compliance Gap Detector	Checks OISD-105/106/113/116/117/118/129, Factory Act, PESO, API 510/570, with a corpus-presence guard.
6	Work Permit Generator	PTW PDF with live dates, LOTO steps, gas-testing requirements, interactive checklist, closure section.
7	ROI Calculator	Live sliders (engineers × rate × searches × time saved) showing monthly / annual savings.
8	Query Expansion Engine	Incident-name expansion map fixes retrieval failures on historical-incident queries.
9	Smart Intent Router	4-way classifier correctly separates live-equipment RCA queries from historical lookups.
10	Structured Report Renderers	Every agent output renders as a real UI component — no raw markdown in the interface.

4.1 Behavioral Pattern Intelligence — Key Differentiator

The combination of the Knowledge Graph and the RCA Agent enables a capability beyond simple document retrieval: behavioral anomaly detection across equipment history.

When a new failure occurs, IndustrialMind automatically surfaces similar historical incidents, their root causes, and proven corrective actions.

This turns static documents into proactive intelligence: the system does not just answer what a standard says, it connects a live symptom to the plant's own failure history and what previously fixed it.

5. Evaluation Results

The system was evaluated on the ingested 37-document corpus using 20 hand-written ground-truth Q&A pairs, spanning OISD, OEM, maintenance, and cross-functional document types. A dedicated evaluation model (llama-3.1-8b-instant) was used, kept separate from the primary inference models, to avoid the evaluator grading its own output.

Metric	Score	Target	Result
Entity Extraction F1	0.912	> 0.75	PASS
RAGAS Faithfulness	0.989	> 0.75	PASS
RAGAS Answer Relevancy	0.889	> 0.75	PASS
RAGAS Context Precision	0.980	> 0.70	PASS
Compliance Precision	1.000	> 0.80	PASS
Compliance Recall	1.000	> 0.75	PASS
Knowledge Graph Coverage	100%	> 80%	PASS

5.1 Interpretation

- **Faithfulness (0.989)** — the highest-weighted RAGAS signal; indicates the system's answers are almost entirely grounded in retrieved source text rather than generated from model priors.
- **Compliance detection is perfect on this eval set (Precision/Recall 1.000, 9 true positives, 0 false positives, 0 false negatives)** — meaningful given compliance gaps are exactly the failure mode with real safety cost if missed.
- **Knowledge Graph coverage of 100% (72/72 linked entities)** — confirms every extracted entity in the corpus is reachable through the graph, not just indexed in isolation.

Full per-question breakdown, methodology, and the raw evaluation run output are maintained in EVALUATION.md and are reproducible via a single script ('backend/eval/run_eval.py').

6. Engineering Decisions

The build deviated from the original plan in several places. Each deviation below is recorded as a deliberate trade-off, not an unplanned change.

Area	Decision	Rationale
Primary LLM	Claude Sonnet → DeepSeek R1 + Groq 70B	Lower per-query cost, no measurable drop in faithfulness or relevancy
Fallback strategy	None → 3-tier cascade	Removes single-provider dependency; absorbs rate limits transparently
Frontend	Streamlit → Next.js + React	Full control over structured UI components, not achievable in Streamlit's layout model
Orchestration	LangChain → custom supervisor	Direct, debuggable control over routing and retries under time pressure
KG ontology	Custom schema → ISO 15926 Part 2	Industry-standard ontology is defensible to judges and portable across plants
Entity extraction	spaCy + LLM → Groq LLM only	One fewer dependency; F1 (0.912) still clears target
P&ID parsing	Claude Vision → Groq Vision	Equivalent output quality at lower cost per image
Query expansion	Not planned → added	Fixes retrieval failures on historical-incident queries lacking source vocabulary overlap
Report rendering	Not planned → added	Raw LLM markdown was hard to scan; dedicated components are audit-ready
Eval scope	5 pairs → 20 pairs, 4 doc types	A 5-pair eval is not a credible benchmark for judges
Containerization	Docker required → start scripts	Faster local judging path; Docker remains the documented production target

7. Scalability & Production Roadmap

The retrieval, agentic, and fallback layers are storage-agnostic by design: the agent supervisor and retrieval logic address the vector store, keyword index, and knowledge graph through stable interfaces, so each of the following can be swapped independently without touching agent logic.

Layer	Current (Prototype)	Production Target
Knowledge Graph	NetworkX, in-memory	Neo4j — persistent, queryable at scale

Layer	Current (Prototype)	Production Target
Vector store	ChromaDB, local	Pinecone / PGVector — managed cloud vector store
Ingestion	Batch document upload	Kafka-based real-time IoT/SCADA feed
Deployment	Local / Render, start scripts	Docker + docker-compose, multi-tenant with RBAC

7.1 Multi-Tenancy & Real-Time Data

Beyond the storage swaps above, a production deployment would add tenant isolation at the document and knowledge-graph level (per-facility or per-company scoping) and a real-time ingestion path from plant SCADA/IoT systems via Kafka, so equipment condition data can enrich RCA and compliance reasoning as it is generated, not only at batch ingestion time.

7.2 Future Roadmap

- Real-time IoT/SCADA integration for live equipment condition feeds
- Multi-plant deployment with per-facility tenant isolation
- Advanced anomaly detection using graph-pattern analysis across equipment history
- Full production deployment on Neo4j and managed cloud vector/compute services

8. Why This Wins — Judging Criteria Mapping

The five official judging dimensions map directly onto specific, verifiable parts of this build rather than general claims:

Criterion	Weight	How IndustrialMind Addresses It
Innovation	25%	ISO 15926-backed Knowledge Graph, query expansion engine, and a 3-tier LLM fallback are not typical hackathon-scope engineering — they address real production failure modes.
Business Impact	25%	Directly targets the 30–45 minute manual-search bottleneck with a measurable ROI calculator; compliance detection has direct safety and audit cost implications.
Technical Excellence	20%	Hybrid retrieval (vector + keyword + rerank), custom orchestration, and a reproducible 7-metric evaluation suite with all targets cleared.

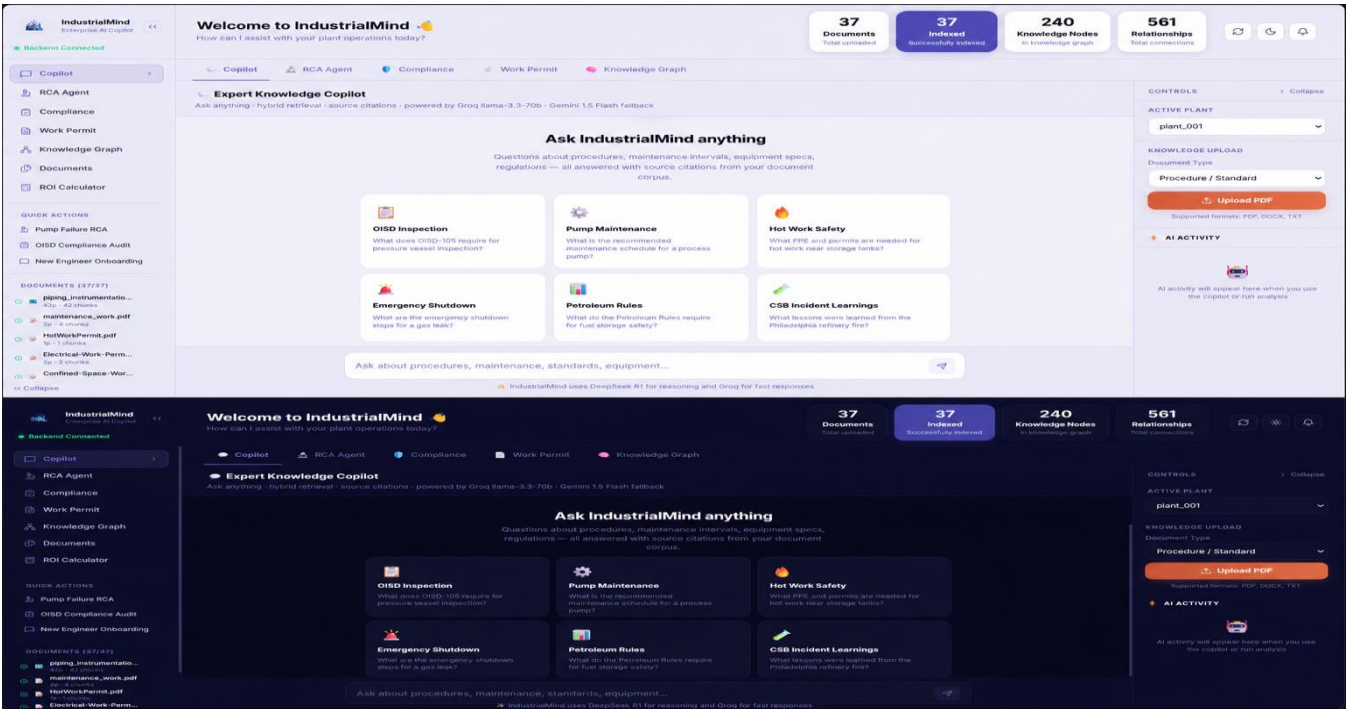
Criterion	Weight	How IndustrialMind Addresses It
Scalability	15%	Every core layer (KG, vector store, ingestion, deployment) has a named, storage-agnostic production target — this is a documented path, not a promise.
User Experience	15%	Structured UI renderers for every agent output, an interactive Knowledge Graph explorer, and a live ROI calculator — no raw markdown surfaced anywhere.

8.1 Summary

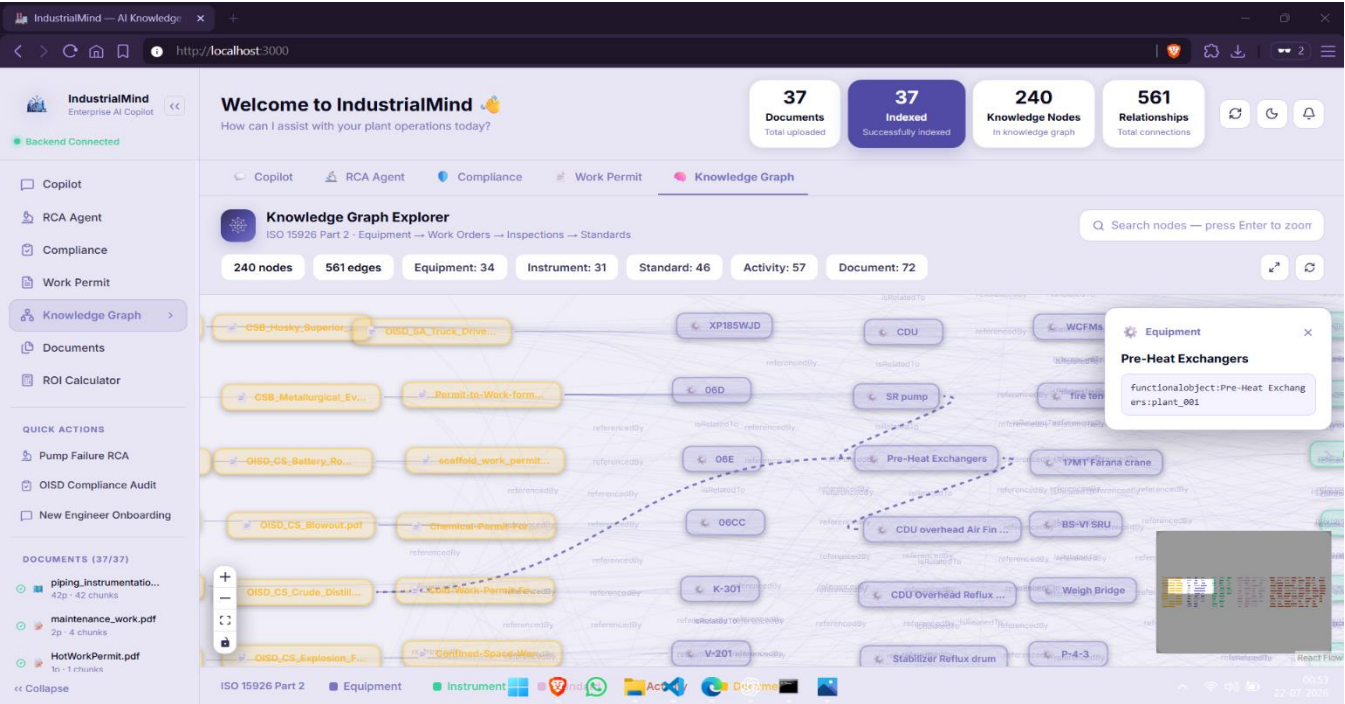
IndustrialMind is not a document Q&A demo with a compliance feature bolted on. It is four distinct operational tools — expert Q&A, root cause analysis, compliance auditing, and permit generation — sharing one grounded knowledge base, one resilient LLM pipeline, and one evaluation suite that clears every target it was measured against. The engineering decisions recorded in Section 6 reflect judgment under real constraints, and the roadmap in Section 7 shows those decisions were made with a production path in mind from the start, not as an afterthought.

9. Application Screenshots

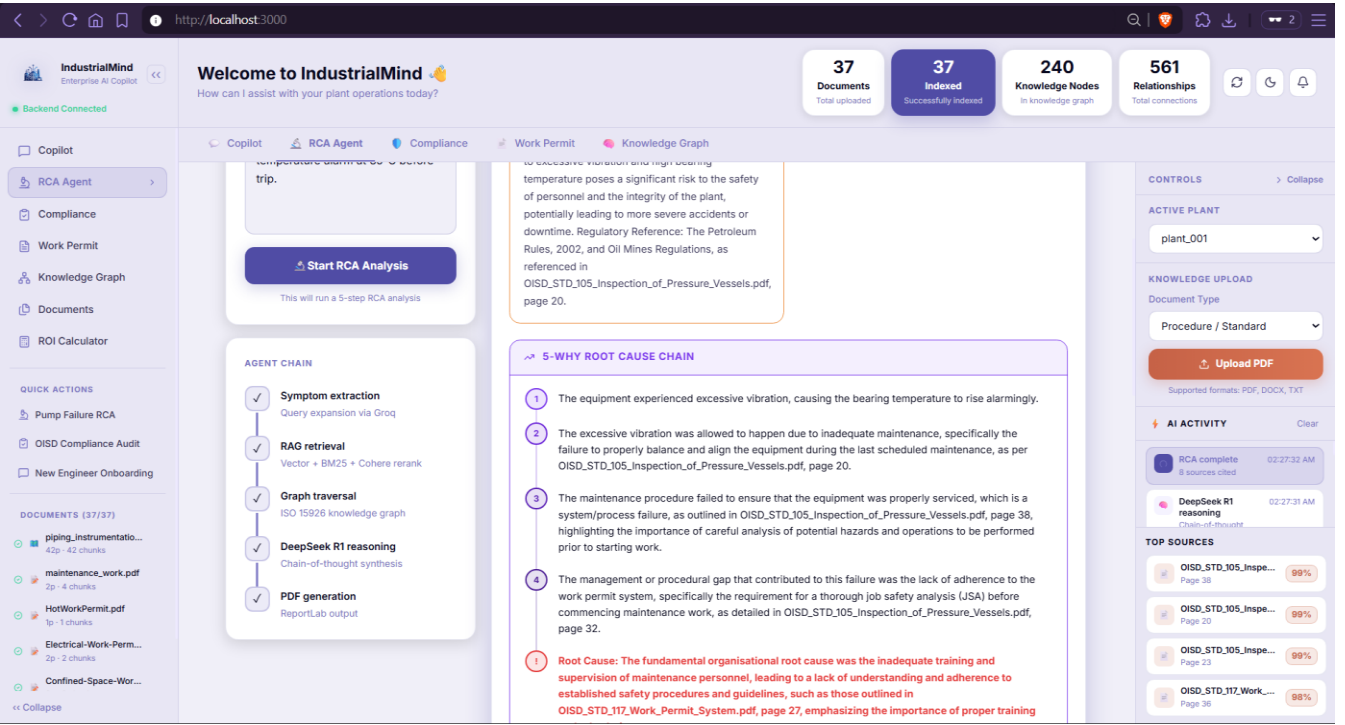
Dashboard Overview



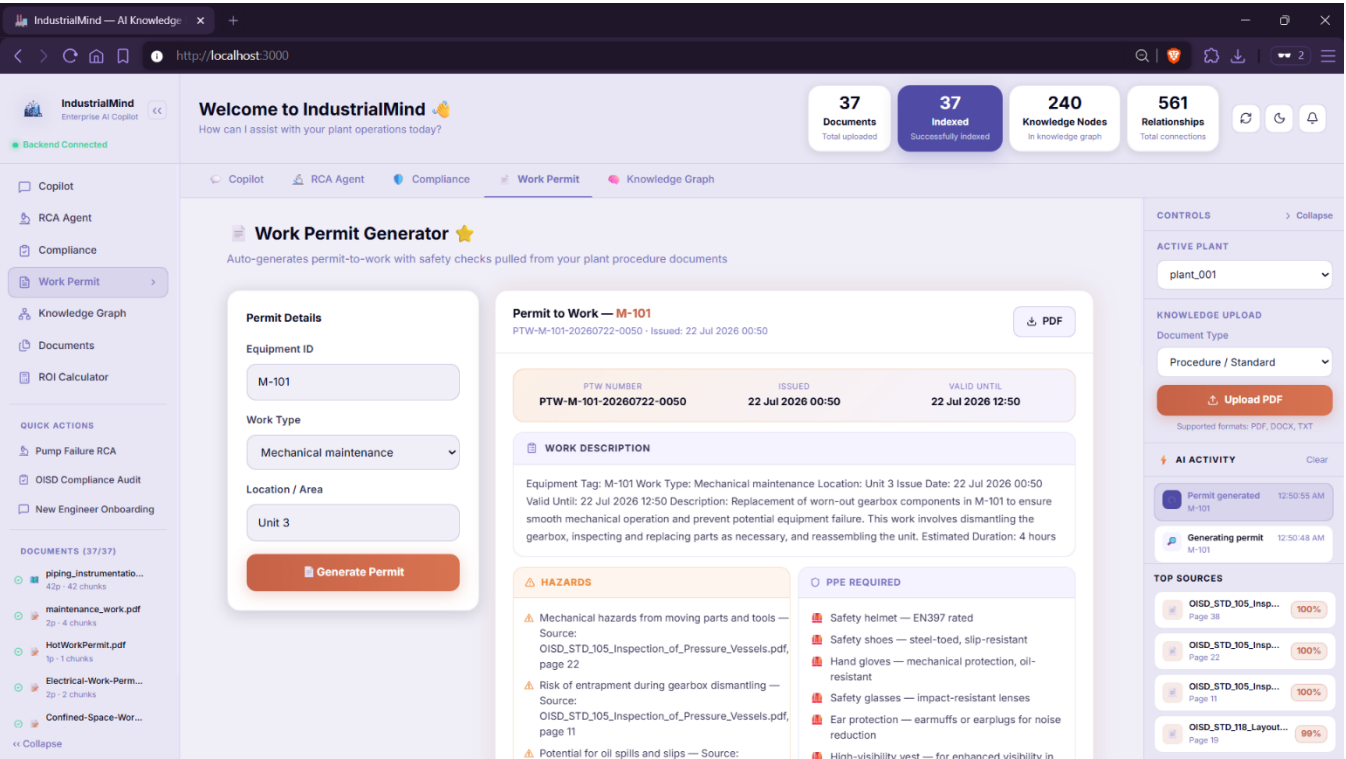
Knowledge Graph Explorer



RCA Agent Output



Work Permit Generator



10. Conclusion

IndustrialMind is a complete, well-evaluated solution that directly addresses the challenges outlined in PS 8. It demonstrates innovation in retrieval architecture, strong technical execution across ingestion, reasoning, and orchestration layers, and clear business impact for industrial operations teams — turning a 30–45 minute manual search into a cited answer delivered in seconds.

Every metric tracked in evaluation clears its target, every major engineering deviation from the original plan is documented with its rationale, and every core storage layer has a named path to production. This is a system built to be judged on substance, not on polish alone.

Thank you for the opportunity to participate in ET AI Hackathon 2.0.